

PREMIUM UNIVERSITY EDITION

ARTIFICIAL INTELLIGENCE MASTER HANDBOOK

Notes · Visuals · Solved Examples · Papers

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- ✓ Beginner → Advanced University Level
 - ✓ Step-by-step explanations with diagrams
 - ✓ Visual learning system with concept maps
 - ✓ 20+ solved university-style papers
 - ✓ Scenario questions & real-world applications
 - ✓ Exam tips, memory tricks & common mistakes

A COMPLETE AI ACADEMY

Teacher · Textbook · Visual Tutor — Combined

FOREWORD — How To Use This Handbook

Welcome, student. This book has been designed for **one promise**: even if you know **absolutely nothing** about Artificial Intelligence today, by the end of this handbook you will be able to **solve any university question independently** — without needing YouTube videos or external lectures.

Every concept is taught in **three layers**:

- **Layer 1 — Real-life story**: We first explain the idea using something you already know (weather, games, recipes, friends).
- **Layer 2 — Technical definition**: Then we give you the proper definition the way universities expect it in exams.
- **Layer 3 — Worked examples & diagrams**: Finally we show you how to solve problems step-by-step with visuals.

TIP: Read each chapter in order. Do not skip the 'Beginner Explanation' section even if it feels easy — it builds the intuition that makes the technical part click.

COMMON MISTAKE: Do **not** try to memorize formulas without first understanding the story. Memorization without understanding fails in scenario questions.

Symbols you will see

Symbol	Meaning
TIP	A shortcut, exam hack, or memory trick
COMMON MISTAKE	An error students make under exam pressure
★ Scenario	A real-world style question — solve it like a story
Q / A	Practice question with full solution

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CHAPTER 1 · GRAPH COLORING PROBLEM

1.1 Beginner Story — The Exam Timetable Problem

Imagine your university has 5 subjects: Math, Physics, AI, English and History. The principal wants to schedule final exams. Rule: **two subjects taken by the same student cannot be scheduled on the same day** — otherwise the student would have to be in two rooms at once.

So the principal draws each subject as a **circle** (called a **vertex**). If two subjects share students, she draws a **line** (called an **edge**) between them. Now she paints each subject with a **color** — and each color represents a different exam day.

The rule of the game: No two connected circles can have the same color. The minimum number of colors she needs = the minimum number of exam days needed. **That is graph coloring.**

This is literally how exam scheduling, classroom timetabling, and even Wi-Fi channel allocation works in the real world.

1.2 Technical Definitions

Graph

A graph $G = (V, E)$ where V is a finite set of **vertices** and E is a set of **edges** (pairs of vertices). Two vertices joined by an edge are called **adjacent**.

Graph Coloring

An assignment of colors to each vertex such that **no two adjacent vertices share the same color**. This is called a **proper coloring**.

Chromatic Number $\chi(G)$

The **minimum** number of colors required to properly color the graph G . Finding $\chi(G)$ is an **NP-Hard** problem in general — meaning no fast algorithm is known for all graphs.

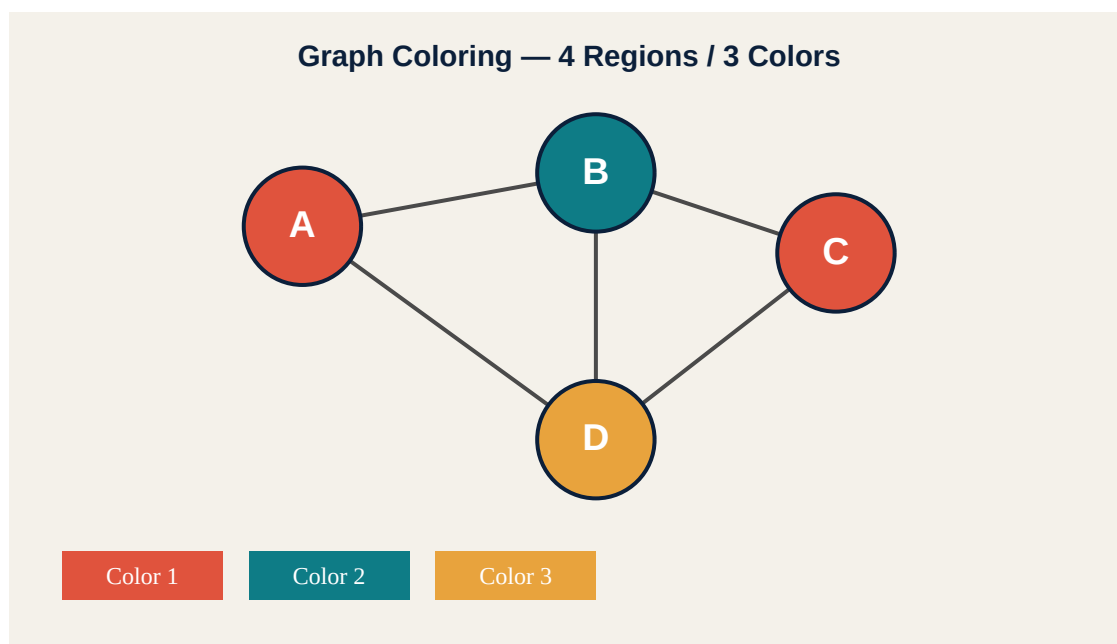


Fig: Proper 3-coloring of a 4-vertex graph. Adjacent vertices have different colors.

1.3 How AI Solves It — Backtracking Algorithm

Graph coloring is a **Constraint Satisfaction Problem (CSP)**. AI uses **backtracking**: try a color for a vertex, move to the next; if a conflict appears, undo the last choice and try a different color.

```

Algorithm GraphColor(G, m):
if all vertices colored: return SUCCESS
pick next vertex v
for color c = 1 to m:
if c is safe for v (no adjacent vertex uses c):
assign color[v] = c
if GraphColor(G, m) == SUCCESS: return SUCCESS
color[v] = 0 // BACKTRACK
return FAILURE
    
```

1.4 Step-By-Step Worked Example

Problem: Color the graph with vertices {A, B, C, D}, edges {(A,B), (A,C), (B,C), (B,D), (C,D)} using minimum colors.

Step 1: Start with vertex A → assign **Red**.

Step 2: Vertex B is adjacent to A → cannot be Red → assign **Green**.

Step 3: Vertex C is adjacent to both A and B → needs a 3rd color → assign **Blue**.

Step 4: Vertex D is adjacent to B (Green) and C (Blue) → can use Red. Assign **Red**.

Result: Minimum colors = 3. Therefore $\chi(G) = 3$.

TIP: Always start with the **highest-degree vertex** (most edges). This 'Welsh-Powell' heuristic usually reaches the chromatic number faster.

1.5 Real-World Applications

Application	What the Color Represents
Exam Timetabling	Exam day / slot
Mobile Tower Frequency	Radio frequency channel
Map Coloring (countries)	Visible color on the map
Sudoku Solving	Digit in each cell
Compiler Register Allocation	CPU register

1.6 Practice — Conceptual + Numerical

Q1. (Easy) Define chromatic number with an example.

Answer: The chromatic number $\chi(G)$ is the minimum number of colors needed for a proper coloring. Example: a triangle K_3 needs $\chi = 3$, because every vertex is adjacent to every other.

Q2. (Medium) Find χ for a complete graph K_n .

Answer: In K_n every vertex is adjacent to every other vertex. Therefore each vertex must have a unique color. $\chi(K_n) = n$. **General rule:** $\chi(K_n) = n$.

Q3. ★ Scenario — University Timetable

A university must schedule 6 exams. Subject-pairs that share students are: (1,2), (1,3), (2,3), (3,4), (4,5), (5,6), (4,6). Find the minimum number of exam days.

Solution: Model each subject as a vertex, shared-student pairs as edges. Sort by degree:

Degrees → 3 has 3 neighbors, 4 has 3, others have 2. Apply Welsh-Powell:

- Color Subject 3 = Red. Subject 4 = Green (not adjacent to 3? Yes adjacent → use Green).
- Subject 1 (adj 2,3=Red) = Green.
- Subject 2 (adj 1=Green, 3=Red) = Blue.
- Subject 5 (adj 4=Green, 6) = Red.
- Subject 6 (adj 4=Green, 5=Red) = Blue.

Result: 3 colors → **3 exam days** suffice.

Q4. (Advanced) Prove that any planar graph is 4-colorable.

Answer (sketch): This is the famous **Four Color Theorem** (Appel & Haken, 1976). Every planar graph (one that can be drawn without crossing edges) can be properly colored using at most 4 colors. Proof relies on showing that the set of 'unavoidable configurations' is reducible — verified by computer-assisted case analysis.

COMMON MISTAKE: Students often confuse 'planar' (drawable without crossings) with 'simple' (no parallel edges). They are *not* the same.

CHAPTER 2 · ALPHA-BETA PRUNING

2.1 Beginner Story — Playing Chess Without Wasting Brain Power

When a chess grandmaster looks at the board, she does **not** calculate every possible move. She quickly realizes: 'If I move my knight here, my opponent will *obviously* capture my queen — so I should not even waste time analyzing the rest of that line.' That instinct of **cutting off useless branches** is exactly what Alpha-Beta Pruning does for an AI game-playing program.

2.2 The Minimax Algorithm (Foundation)

Two players: **MAX** (wants to maximize score) and **MIN** (wants to minimize MAX's score). The game tree alternates between MAX nodes and MIN nodes. Minimax recursively computes the value of each node as:

```
value(MAX node) = max(values of children)
value(MIN node) = min(values of children)
```

2.3 Alpha-Beta Pruning — The Optimization

Alpha-Beta keeps two running bounds while exploring:

- α (**alpha**) — the best score MAX can *guarantee so far*. Starts at $-\infty$.
- β (**beta**) — the best score MIN can *guarantee so far*. Starts at $+\infty$.

Pruning Rule: Whenever $\alpha \geq \beta$, stop exploring this branch. It cannot affect the final decision.

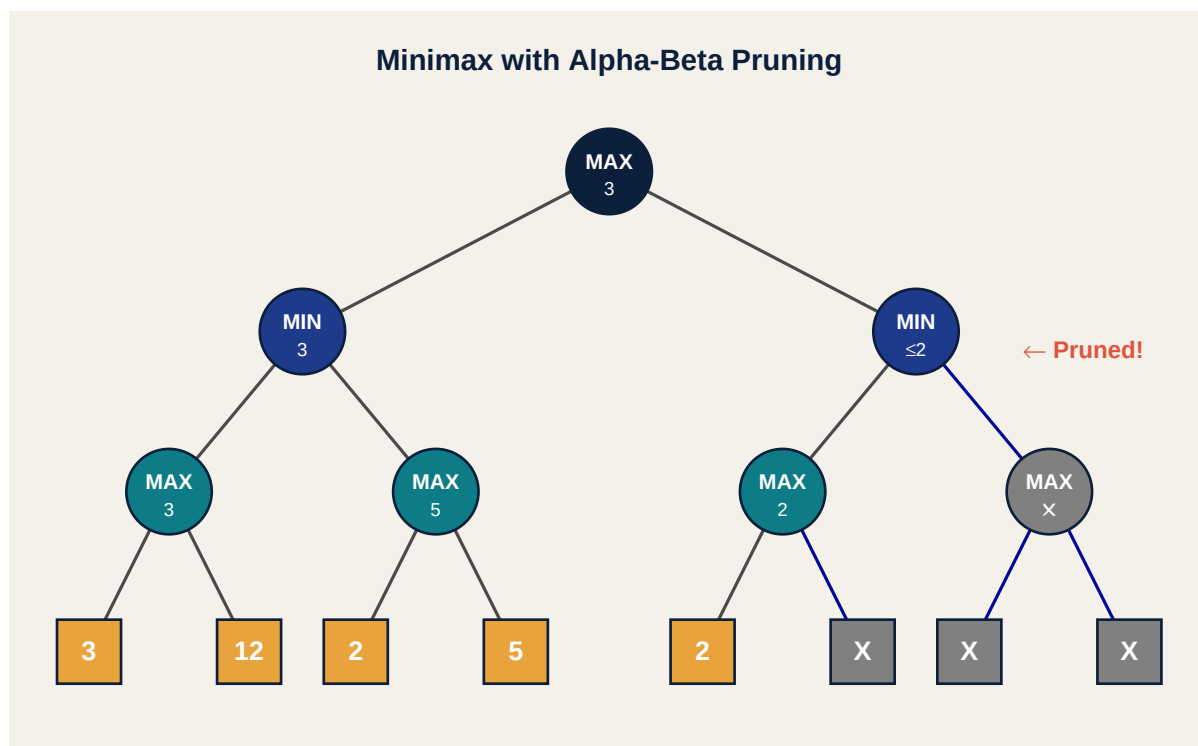


Fig: Minimax tree with Alpha-Beta pruning. The right subtree below the second MIN node is pruned ($\leq 2 \leq \alpha=3$).

2.4 Step-by-Step Worked Example

Tree leaves left→right: 3, 12, 2, 5, 2, 14, 1, 8. Maximize at root.

Step 1: Explore leftmost MIN node. Children = {3, 12}. $\min(3, 12) = 3$. Now $\alpha=3$ at root MAX.

Step 2: Next MIN node. First child of its MAX subnode = 2. So this $\text{MIN} \leq 2$.

Step 3: Because $2 < \alpha$ (3), root MAX will never choose this path. **Prune** the rest of the subtree.

Step 4: Continue with remaining branches the same way. Root value = 3.

TIP: Order children **best-first**. With perfect ordering, alpha-beta examines only $O(b^{(d/2)})$ nodes instead of $O(b^d)$ — effectively **doubling** your search depth.

2.5 Complexity Comparison

Algorithm	Time	Best Case Time	Practical Depth
Minimax	$O(b^d)$	$O(b^d)$	Shallow
Alpha-Beta (random order)	$O(b^d)$	$O(b^{(3d/4)})$	Medium
Alpha-Beta (best order)	$O(b^{(d/2)})$	$O(b^{(d/2)})$	$\approx 2\times$ deeper

2.6 Applications

- Chess engines (Stockfish, Deep Blue lineage)
- Othello / Reversi AI
- Connect-4 perfect players
- Tic-Tac-Toe AI tutorials
- Any 2-player zero-sum game

2.7 Practice Questions

Q1. Why does Alpha-Beta produce the same answer as Minimax?

Answer: Pruning only removes branches that *cannot influence* the final value (because a better choice already exists). The chosen root value is therefore identical to plain Minimax — only the work done is less.

Q2. ★ Scenario — Order Matters

A MAX-MIN-MAX tree has 8 leaves: [3, 5, 6, 9, 1, 2, 0, -1]. Show how many leaves Alpha-Beta examines if (a) left-to-right and (b) sorted best-first for MAX.

(a) Standard L→R: ≈ 6 of 8 leaves examined. (b) Best-first ordering: only 4 leaves examined. This demonstrates why *move ordering* is a huge optimization in real engines.

COMMON MISTAKE: Many students prune *after* reading a node's value. The correct rule: check $\alpha \geq \beta$ **before** exploring the next child.

CHAPTER 3 · EXPERT SYSTEMS

3.1 Beginner Story — Bottling a Doctor's Brain

Imagine you could record every decision Dr. Khan, a famous cardiologist, makes when diagnosing patients. You could store all his rules of thumb ('if chest pain + shortness of breath + age > 50 → suspect heart attack') in a computer. Now anyone in a remote village without a doctor could enter symptoms and get a near-expert diagnosis. That stored-doctor program is an **Expert System**.

3.2 Definition

An Expert System is an AI program that **emulates the decision-making ability of a human expert** in a narrow domain, using a **knowledge base** of rules and an **inference engine** that reasons over those rules.

3.3 Architecture

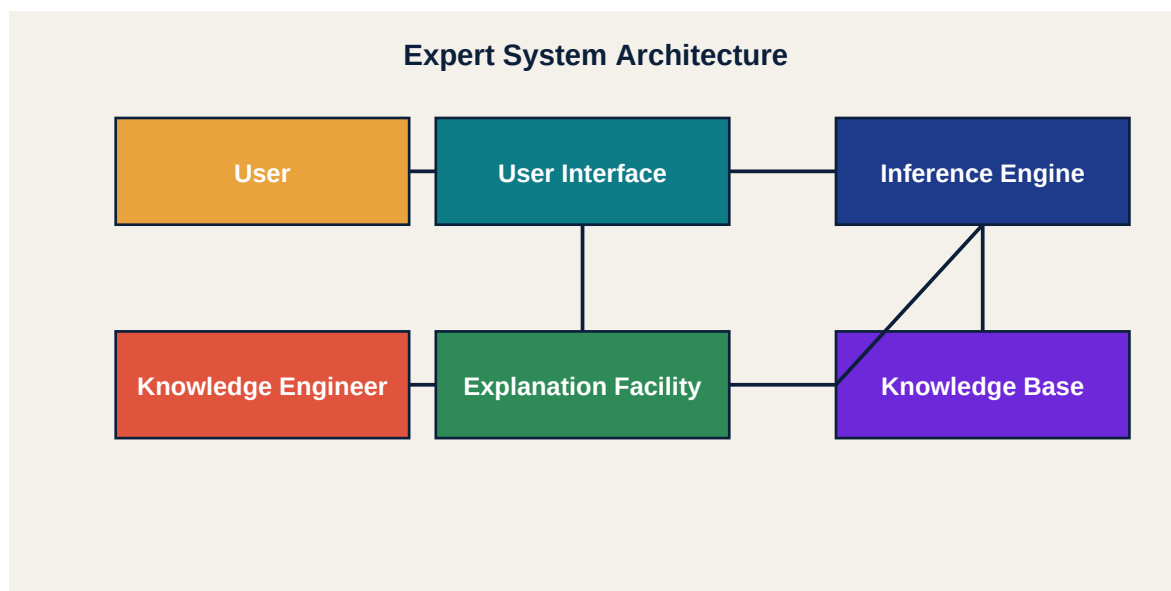


Fig: Six-component architecture: User, UI, Inference Engine, Knowledge Base, Explanation Facility, Knowledge Engineer.

Components Explained

Component	Job
Knowledge Base	Stores facts and IF-THEN rules from the human expert.
Inference Engine	Applies rules to facts to derive new conclusions.
User Interface	Lets non-experts ask questions in plain language.
Explanation Facility	Tells the user WHY a conclusion was reached.
Knowledge Engineer	Human who extracts knowledge from the expert.
Working Memory	Temporary scratchpad of currently known facts.

3.4 Two Reasoning Strategies

Forward Chaining (Data-Driven)

Start with the facts you have. Apply rules to derive new facts. Repeat until you reach a goal or no more rules fire.

Facts: { fever, cough }
Rule R1: IF fever AND cough THEN suspect_flu
Apply R1 → add suspect_flu
Rule R2: IF suspect_flu AND body_ache THEN diagnose_flu → not yet

Backward Chaining (Goal-Driven)

Start with a hypothesis (goal). Ask: 'What facts would prove this?' Work backward through the rules.

Goal: diagnose_flu?
Need: suspect_flu AND body_ache
To prove suspect_flu, need fever AND cough
Ask user: do you have fever? cough? body_ache?

3.5 Famous Real Expert Systems

System	Domain	Year
DENDRAL	Chemistry (molecular structure)	1965
MYCIN	Medical (bacterial infections)	1972
XCON	Computer configuration (DEC)	1980
PROSPECTOR	Mineral exploration	1978
CADUCEUS	Internal medicine	1984

3.6 Advantages & Limitations

■ Advantages	■ Limitations
Available 24/7, never tires	Cannot handle situations outside its rules
Consistent decisions	No common sense or intuition
Captures rare expertise	Knowledge acquisition is slow
Can explain its reasoning	Cannot learn from new cases without updates

3.7 Practice

Q1. ★ Scenario — Build a Mini Medical Expert System

Design 3 rules to diagnose 'common cold' vs 'flu' vs 'allergy'.

Answer:

R1: IF runny_nose AND no_fever AND itchy_eyes THEN allergy
R2: IF runny_nose AND mild_fever AND sore_throat THEN common_cold
R3: IF high_fever AND body_ache AND fatigue THEN flu

Forward chain when user enters symptoms; backward chain when user asks 'Do I have flu?'

TIP: In exams, always draw the 6-box architecture diagram. It earns easy marks.

CHAPTER 4 · NATURAL LANGUAGE PROCESSING (NLP)

4.1 Beginner Story — Teaching a Computer to Read

When you read 'I love eating mangoes', your brain instantly knows: *I* = the speaker, *love* = a feeling, *mangoes* = a fruit, and the overall sentiment is positive. Computers do not have that intuition. NLP is the branch of AI that **teaches machines to understand and generate human language** — exactly the technology behind ChatGPT, Google Translate, Siri and Alexa.

4.2 The NLP Pipeline

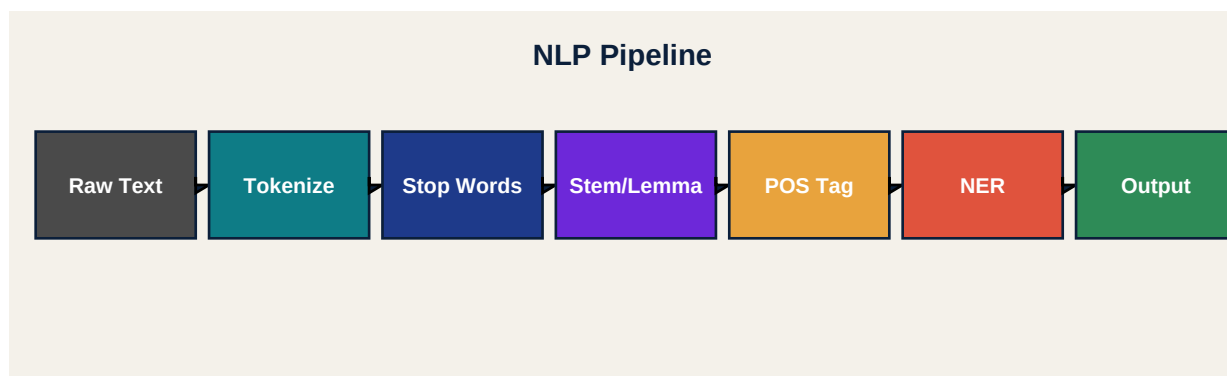


Fig: Standard NLP processing pipeline: raw text → tokens → cleaned → analyzed → labeled → output.

Stage 1 — Tokenization

Break the sentence into **tokens** (words or sub-words).

Input : 'The cats are running.'
Output: ['The', 'cats', 'are', 'running', '.']

Stage 2 — Stop Word Removal

Remove very common words ('the', 'is', 'a') that carry little meaning.

After removal: ['cats', 'running']

Stage 3 — Stemming vs Lemmatization

Stemming chops word endings crudely (running → run, studies → studi). **Lemmatization** uses a dictionary to return real base words (studies → study, better → good).

Stage 4 — POS Tagging

Label each token with its **part of speech**: noun (NN), verb (VB), adjective (JJ), etc.

'cats/NN run/VB fast/RB'

Stage 5 — Named Entity Recognition (NER)

Detect proper nouns: persons, organizations, locations, dates.

'Elon Musk founded SpaceX in 2002.'
→ [Elon Musk]=PERSON, [SpaceX]=ORG, [2002]=DATE

Stage 6 — Sentiment Analysis

Classify text as positive / negative / neutral. Used heavily in product reviews and social-media monitoring.

4.3 Real-World Applications

- **ChatGPT / Claude / Gemini** — large language models
- **Google Translate** — neural machine translation
- **Voice Assistants** — Siri, Alexa, Google Assistant
- **Spam filters** in Gmail / Outlook
- **Auto-complete & smart reply** in keyboards
- **Resume screening** by HR software

4.4 Practice

Q1. Tokenize and lemmatize: 'The children were playing happily in the gardens.'

Tokens: [The, children, were, playing, happily, in, the, gardens]

After stop-word removal: [children, playing, happily, gardens]

Lemmas: [child, play, happy, garden]

Q2. ★ Scenario — Sentiment of a Movie Review

Classify: 'The plot was predictable but the acting was outstanding and the music was incredible.'

Solution: Identify polarity words → 'predictable' (slightly negative), 'outstanding' (strong positive), 'incredible' (strong positive). Sum of weighted polarities is positive. **Overall: POSITIVE.**

COMMON MISTAKE: Negations flip polarity. 'Not good' is negative even though 'good' is positive. Always handle 'not' / 'never' explicitly.

CHAPTER 5 · N-GRAM LANGUAGE MODEL

5.1 Beginner Story — How Your Phone Predicts the Next Word

When you type 'I am going to the...' your phone suggests 'office', 'market', 'gym'. How does it know? It has counted, across millions of sentences, which word usually follows 'going to the'. That counting trick is an **N-Gram model**.

5.2 Definition

An **N-gram** is a contiguous sequence of N items (words). The model estimates the probability of a word based on the previous N-1 words:

$$P(w_{\blacksquare} \mid w_{\blacksquare}, \dots, w_{\blacksquare\blacksquare\blacksquare}) \approx P(w_{\blacksquare} \mid w_{\blacksquare\blacksquare\blacksquare\blacksquare}, \dots, w_{\blacksquare\blacksquare\blacksquare})$$

N-Gram Models

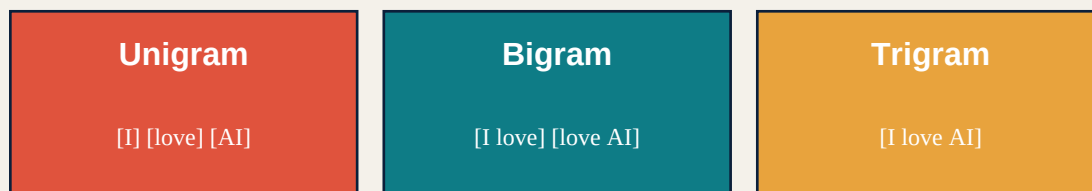


Fig: Unigram / Bigram / Trigram segmentation of the sentence 'I love AI'.

5.3 Probability Formulas

Model	Formula	Context Used
Unigram	$P(w)$	none
Bigram	$P(w_{\blacksquare} \mid w_{\blacksquare\blacksquare}) = \text{Count}(w_{\blacksquare\blacksquare}, w_{\blacksquare}) / \text{Count}(w_{\blacksquare\blacksquare})$	1 previous word
Trigram	$P(w_{\blacksquare} \mid w_{\blacksquare\blacksquare}, w_{\blacksquare\blacksquare\blacksquare})$	2 previous words

5.4 Worked Example

Corpus: 'I love AI. I love Python. I study AI.'

Count(I) = 3, Count(love) = 2, Count(AI) = 2, Count(I love) = 2, Count(love AI) = 1.

Bigram $P(\text{love} \mid \text{I}) = \text{Count}(\text{I love}) / \text{Count}(\text{I}) = 2/3 \approx 0.67$.

Bigram $P(\text{AI} \mid \text{love}) = \text{Count}(\text{love AI}) / \text{Count}(\text{love}) = 1/2 = 0.50$.

5.5 Smoothing (Laplace / Add-One)

Problem: an unseen bigram gives probability 0, killing the entire sentence probability. Fix with Laplace smoothing:

$$P_{\text{smooth}}(w_{\blacksquare} \mid w_{\blacksquare\blacksquare}) = (\text{Count}(w_{\blacksquare\blacksquare}, w_{\blacksquare}) + 1) / (\text{Count}(w_{\blacksquare\blacksquare}) + V)$$

where V = vocabulary size

5.6 Applications

- Mobile keyboard next-word prediction
- Search-engine query auto-complete
- Spell checking & grammar correction
- Speech recognition language scoring
- Pre-deep-learning machine translation

TIP: Bigrams capture local context cheaply. Trigrams are more accurate but explode in storage — vocabulary V means V^3 possible trigrams.

CHAPTER 6 · MARKOV CHAIN

6.1 Beginner Story — Tomorrow's Weather

Suppose tomorrow's weather depends **only** on today's weather — not on the whole past month. If today is Sunny, there is a 70% chance tomorrow is Sunny and 30% chance it is Rainy. This 'only-yesterday-matters' rule is the **Markov Property**.

6.2 Definition

A Markov Chain is a sequence of random variables $X_1, X_2, X_3 \dots$ taking values in a finite set of **states**, such that:

$$P(X_{n+1} = s \mid X_1, X_2, \dots, X_n) = P(X_{n+1} = s \mid X_n)$$

The future is independent of the past, given the present.

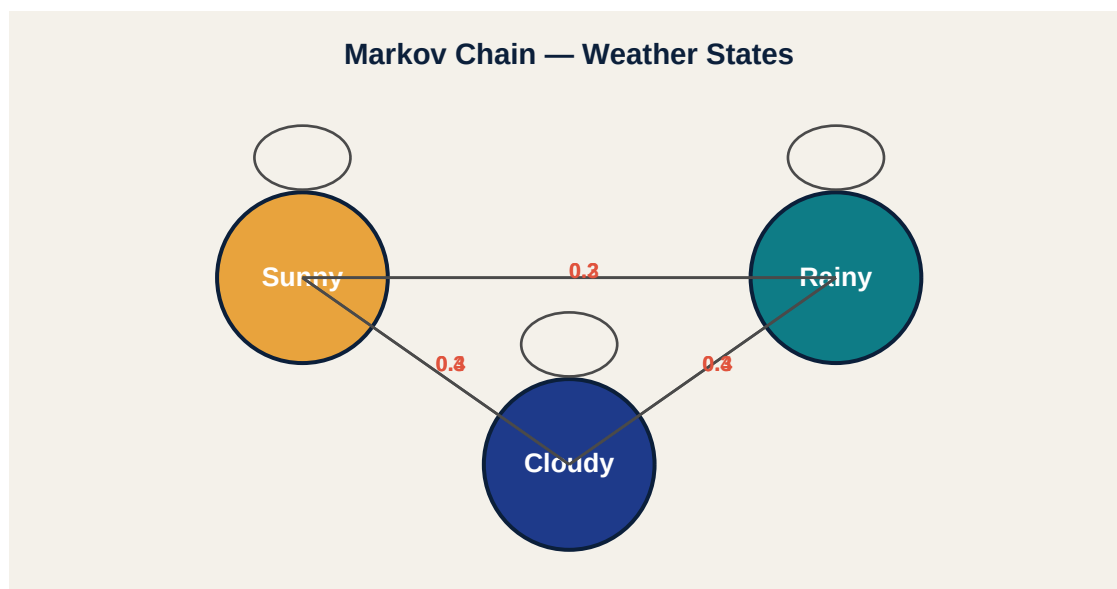


Fig: 3-state weather Markov chain with labeled transition probabilities.

6.3 Transition Matrix

All transitions can be packed into a square matrix P where $P[i][j]$ = probability of going from state i to state j . Rows must sum to 1.

From \ To	Sunny	Rainy	Cloudy
Sunny	0.5	0.2	0.3
Rainy	0.3	0.3	0.4
Cloudy	0.4	0.3	0.3

6.4 Worked Problem

Today is Sunny. What is the probability it will be Rainy **day after tomorrow**?

Step 1: Compute P^2 (matrix square). Entry (Sunny, Rainy) of $P^2 = \sum P(\text{Sunny} \rightarrow k) \cdot P(k \rightarrow \text{Rainy})$.

$$= 0.5 \cdot 0.2 + 0.2 \cdot 0.3 + 0.3 \cdot 0.3 = 0.10 + 0.06 + 0.09 = \mathbf{0.25}.$$

Answer: 25% chance of Rainy day after tomorrow.

6.5 Steady State

As $n \rightarrow \infty$, the probability distribution often converges to a unique **steady-state vector** π satisfying $\pi P = \pi$. This represents the long-run fraction of time spent in each state.

6.6 Applications

- PageRank algorithm (Google search)
- Weather prediction
- Customer behavior modelling (churn)
- Board games — e.g. Snakes & Ladders
- Genetics — DNA sequence modelling

CHAPTER 7 · HIDDEN MARKOV MODEL (HMM)

7.1 Beginner Story — Guessing the Weather From Someone's Umbrella

You are locked in a windowless room. Each day a friend walks in — sometimes with an umbrella, sometimes without. You cannot see the weather (it is **hidden**), but you can *observe* the umbrella. From those observations you want to guess the actual weather sequence. That is exactly what an HMM solves.

7.2 Components

Symbol	Meaning
States S	Hidden states (e.g., Sunny, Rainy)
Observations O	Visible outputs (e.g., Umbrella, No Umbrella)
A	Transition probabilities between hidden states
B	Emission probabilities $P(\text{observation} \mid \text{state})$
π	Initial state distribution

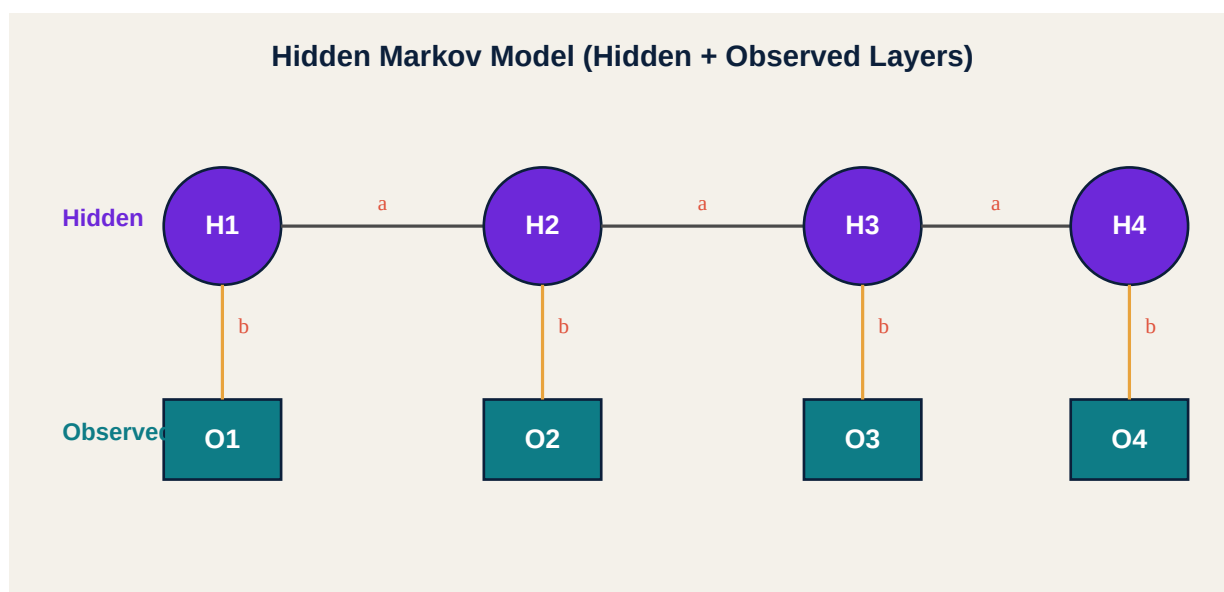


Fig: Two-layer HMM: hidden states $H1 \dots H4$ emit observations $O1 \dots O4$. A = transition, B = emission.

7.3 The Three Classic Problems

Problem	Question	Algorithm
Evaluation	Given model λ and observation sequence O , find $P(O \lambda)$	Forward Algorithm
Decoding	Most likely hidden state sequence given O	Viterbi Algorithm
Learning	Estimate λ from data	Baum–Welch (EM)

7.4 Mini Numerical (Forward Algorithm Idea)

States: {Rainy, Sunny}. Observations: {Umbrella, NoUmbrella}.

$\pi = [\text{Rainy}=0.6, \text{Sunny}=0.4]$. $A = [[0.7, 0.3], [0.4, 0.6]]$. $B = [[U=0.9, N=0.1], [U=0.2, N=0.8]]$.

Observation Day 1 = Umbrella. $\alpha_{\text{Rainy}}=0.6 \cdot 0.9=0.54$, $\alpha_{\text{Sunny}}=0.4 \cdot 0.2=0.08$.

Therefore the model already strongly prefers **Rainy** on Day 1 given the umbrella sighting.

7.5 Applications

- Speech recognition (phoneme decoding)
- Part-of-speech tagging in NLP
- Bioinformatics — gene finding, protein structure
- Handwriting & gesture recognition
- Financial regime detection

COMMON MISTAKE: Students often confuse Markov Chain and HMM. **Key difference:** in a Markov Chain, states are *directly observed*. In an HMM, states are *hidden*; only emissions are observed.

CHAPTER 8 · PROPOSITIONAL LOGIC

8.1 Beginner Story — Yes / No Statements

Every sentence in propositional logic is either **True** or **False** — no in-between. 'It is raining' is a proposition. 'What time is it?' is not (it has no truth value).

8.2 Connectives

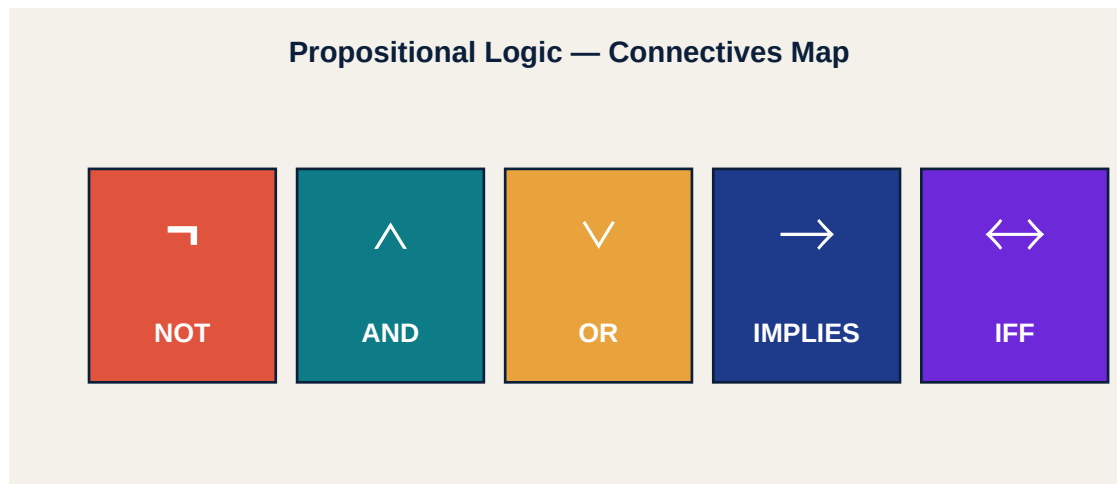


Fig: The five fundamental logical connectives.

Connective	Symbol	Meaning
Negation	$\neg P$	NOT P
Conjunction	$P \wedge Q$	P AND Q
Disjunction	$P \vee Q$	P OR Q
Implication	$P \rightarrow Q$	If P then Q
Biconditional	$P \leftrightarrow Q$	P if and only if Q

8.3 Truth Table — Implication

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Implication is FALSE only when the antecedent is TRUE but the consequent is FALSE. A false promise is only broken when the condition was actually met.

8.4 Rules of Inference

Rule	Form	Example
Modus Ponens	$P, P \rightarrow Q \blacksquare Q$	Rain, Rain $\rightarrow$ Wet $\blacksquare$ Wet
Modus Tollens	$\neg Q, P \rightarrow Q \blacksquare \neg P$	Not Wet, Rain $\rightarrow$ Wet $\blacksquare$ Not Rain

Hypothetical Syll.	$P \rightarrow Q, Q \rightarrow R \blacksquare P \rightarrow R$	Chain reasoning
Disjunctive Syll.	$P \vee Q, \neg P \blacksquare Q$	Either tea or coffee, not tea, so coffee
Resolution	$P \vee Q, \neg P \vee R \blacksquare Q \vee R$	Used in AI theorem proving

8.5 Worked Example — Prove Wet via Modus Ponens

Premises: (1) 'If it rains then the ground is wet' = $R \rightarrow W$. (2) 'It rains' = R .

Apply Modus Ponens $\rightarrow$ derive W ('the ground is wet'). ■

8.6 Practice

Q. Show that $(P \rightarrow Q) \wedge (Q \rightarrow R) \wedge P$ implies R .

Step 1: From P and $P \rightarrow Q$ (Modus Ponens) $\rightarrow Q$.

Step 2: From Q and $Q \rightarrow R$ (Modus Ponens) $\rightarrow R$.

Hence R is derived. ■

CHAPTER 9 · PREDICATE LOGIC (FIRST-ORDER LOGIC)

9.1 Beginner Story — When Propositions Are Not Enough

Propositional logic can say 'Socrates is mortal'. But it cannot say 'EVERY human is mortal' or 'SOME bird can fly'. We need **variables** and **quantifiers**. That is predicate logic.

9.2 Building Blocks

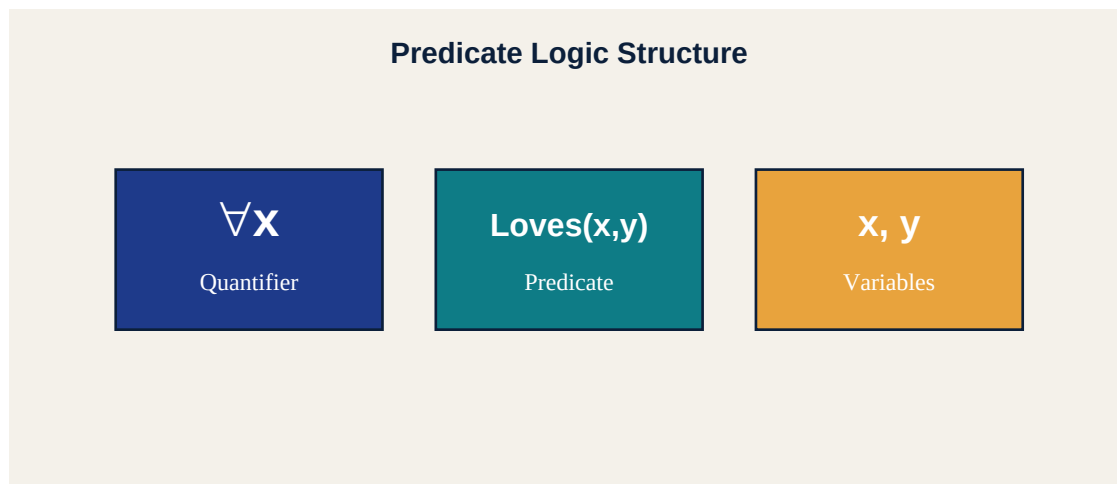


Fig: Predicate logic structure: quantifier + predicate + variables.

- **Constants** — specific objects (Socrates, 5, Paris)
- **Variables** — placeholders (x, y, z)
- **Predicates** — properties or relations: $\text{Mortal}(x)$, $\text{Loves}(x, y)$
- **Functions** — return objects: $\text{FatherOf}(x)$
- **Quantifiers** — $\forall$ (for all), $\exists$ (there exists)

9.3 English → Logic Translation

English	Predicate Logic
All humans are mortal.	$\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$
Some birds can fly.	$\exists x (\text{Bird}(x) \wedge \text{CanFly}(x))$
No fish is a mammal.	$\forall x (\text{Fish}(x) \rightarrow \neg \text{Mammal}(x))$
Every student loves some subject.	$\forall x \exists y (\text{Student}(x) \rightarrow \text{Loves}(x,y))$
There is a student who likes every subject.	$\exists x \forall y (\text{Student}(x) \wedge \text{Subject}(y) \rightarrow \text{Likes}(x,y))$

9.4 Quantifier Negation Rules

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$
$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

'Not everyone is happy' = 'Someone is not happy'.

9.5 Famous Example — The Socrates Syllogism

Premises:

- All humans are mortal: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$
- Socrates is human: $\text{Human}(\text{Socrates})$

Apply Universal Instantiation (substitute $x = \text{Socrates}$): $\text{Human}(\text{Socrates}) \rightarrow \text{Mortal}(\text{Socrates})$.

Then Modus Ponens with premise 2 $\rightarrow$ **Mortal(Socrates)**. ■

9.6 Practice

Q. Translate: 'Every dog has an owner who loves it.'

Answer: $\forall x (\text{Dog}(x) \rightarrow \exists y (\text{Owner}(y, x) \wedge \text{Loves}(y, x)))$

COMMON MISTAKE: Order of quantifiers matters. $\forall x \exists y \text{ Loves}(x,y)$ ('everyone has someone they love') is NOT the same as $\exists y \forall x \text{ Loves}(x,y)$ ('there is one person everyone loves').

MASTER PRACTICE — MCQs · SHORT · LONG · SCENARIOS

Part A — 30 MCQs With Answers

#	Question	A	B	C	D	Ans
1	Chromatic number of K_n is	3	4	5	2	B
2	Alpha-Beta pruning is an optimization of	BFS	DFS	Minimax	A*	C
3	Forward chaining is	Data-driven	Goal-driven	Random	Statistical	A
4	MYCIN is famous as a(n)	Search engine	Expert system	Chatbot	Compiler	B
5	Tokenization belongs to	NLP	Vision	Robotics	Logic	A
6	In a bigram model, context length is	0	1	2	3	B
7	Markov property says future depends on	All past	Present only	Future	Random	B
8	HMM observations are	Hidden	Visible	Partial	Random	B
9	Modus Ponens form is	$P, P \rightarrow Q \vdash Q$	$P, P \rightarrow Q \vdash \neg P$	$P \vee Q, \neg P \vdash Q$	$P \wedge Q \vdash P$	A
10	$\forall x \neg P(x)$ is equivalent to	$\neg \exists x P(x)$	$\exists x \neg P(x)$	$\neg P(x)$	$\forall x P(x)$	A
11	Welsh-Powell algorithm orders vertices by	Name	Degree	Color	Random	B
12	Best alpha-beta complexity is	$O(b^d)$	$O(b^{(d/2)})$	$O(d^2)$	$O(\log d)$	B
13	Inference engine uses	Database	Rules	Pixels	Audio	B
14	Lemmatization differs from stemming by using	Dictionary	Pixels	Audio	RNG	A
15	Trigram needs	Last word	Last 2 words	Last 3 words	All words	B
16	PageRank uses	HMM	Markov Chain	NLP	Logic	B
17	Viterbi solves HMM's	Evaluation	Decoding	Learning	All	B
18	$\neg(P \wedge Q)$ equals	$\neg P \wedge \neg Q$	$\neg P \vee \neg Q$	$P \vee Q$	$P \rightarrow Q$	B
19	Quantifier $\exists$ means	For all	There exists	Not	Implies	B
20	4-Color Theorem applies to	Trees	Planar graphs	Trees only	Random graphs	B
21	In game tree, leaf values come from	Random	Evaluation function	User	DB	B
22	Backward chaining starts from	Facts	Goal	Random rule	UI	B
23	POS tag NN means	Adjective	Noun	Verb	Adverb	B
24	Smoothing fixes	Memory	Zero probability	Speed	Tokens	B
25	Markov matrix rows sum to	0	1	∞	V	B
26	In HMM, B matrix is	Transition	Emission	Initial	Final	B
27	Resolution is used in	NLP	Theorem proving	Vision	Audio	B
28	All birds fly is best written as	$\forall x \text{ Bird}(x) \rightarrow \text{Fly}(x)$	$\forall x \text{ Bird}(x)$	$\text{Fly}(x)$	$\forall x \text{ Fly}(x)$	A
29	Sudoku can be modeled as	NLP	Graph coloring	HMM	Markov	B
30	Stockfish uses	HMM	Alpha-Beta	NLP	CSP	B

Part B — 10 Short Answer Questions (2-3 marks)

Q. Define chromatic number.

A. Minimum number of colors needed to properly color a graph G so that no two adjacent vertices share a color.

Q. State the Markov property.

A. The conditional distribution of the future state depends only on the present state, not on past history.

Q. Differentiate stemming vs lemmatization.

A. Stemming chops endings using rules; lemmatization returns dictionary-valid base forms.

Q. What is α (alpha) in pruning?

A. The best value MAX can guarantee so far on the path to the root.

Q. Two roles of the inference engine?

A. (i) Match rules against facts; (ii) Derive new facts (forward) or prove goals (backward).

Q. Why is smoothing needed in N-grams?

A. To avoid zero probability for unseen word combinations which would zero out entire sentence probabilities.

Q. Define emission probability.

A. Probability of observing a particular output symbol from a hidden state in an HMM, $P(O|S)$.

Q. Modus Tollens rule?

A. From $\neg Q$ and $P \rightarrow Q$ infer $\neg P$.

Q. Difference between $\forall$ and $\exists$?

A. $\forall$ asserts a property holds for every element; $\exists$ asserts at least one element has it.

Q. Why is graph coloring NP-Hard?

A. No polynomial-time algorithm is known to find $\chi(G)$ for arbitrary graphs; verification is easy but search is exponential.

SOLVED UNIVERSITY PAPERS

Each paper carries **50 marks** · **5 questions** · **10 marks each**. Solve under timed conditions (90 min) for best results.

PAPER 1 · 50 Marks · Time: 90 min

Q1. (10 marks) Define graph coloring. Prove that $\chi(K_n) = n$. Apply backtracking to color the graph with vertices $\{A,B,C,D,E\}$ and edges $\{(A,B),(A,C),(B,C),(B,D),(C,E),(D,E)\}$.

Solution: Graph coloring assigns colors to vertices so adjacent ones differ. In K_n every pair is adjacent $\rightarrow$ all vertices need different colors $\rightarrow \chi(K_n)=n$. For the given graph: A=Red, B (adj A) =Green, C (adj A,B)=Blue, D (adj B)=Red, E (adj C,D)=Green. $\chi=3$.

Q2. (10 marks) Explain the Minimax algorithm. Apply alpha-beta pruning on the tree with leaves $[3,5,6,9,1,2,0,-1]$ (depth 3, $b=2$). Show pruned branches.

Solution: Minimax: MAX maximizes, MIN minimizes alternately. After exploring left subtree we get value 5; root $\alpha=5$. Right MIN subtree's first MAX child yields 2, so $\text{MIN} \leq 2 < \alpha=5 \rightarrow$ prune the remaining MAX subtree. Root value = 5.

Q3. (10 marks) Draw and explain the architecture of an expert system. Differentiate forward and backward chaining with examples.

Solution: Architecture includes UI, Inference Engine, Knowledge Base, Explanation Facility, Working Memory, Knowledge Engineer. Forward chaining is data-driven: facts trigger rules until a goal is reached (used in MYCIN diagnosis). Backward chaining is goal-driven: starting from a hypothesis, sub-goals are proved (used in PROLOG).

Q4. (10 marks) Tokenize, remove stop words and lemmatize: 'The autonomous cars were navigating the busy streets carefully.' Then identify named entities.

Solution: Tokens: [The, autonomous, cars, were, navigating, the, busy, streets, carefully]. Stop-word removed: [autonomous, cars, navigating, busy, streets, carefully]. Lemmas: [autonomous, car, navigate, busy, street, careful]. NER: none (no proper nouns).

Q5. (10 marks) Given a corpus with $\text{Count}(I \text{ love})=4$, $\text{Count}(I)=10$, $\text{Count}(\text{love } AI)=3$, $\text{Count}(\text{love})=6$, vocabulary $V=1000$. Compute bigram probabilities $P(\text{love}|I)$ and $P(AI|\text{love})$ with and without Laplace smoothing.

Solution: Without smoothing: $P(\text{love}|I)=4/10=0.4$, $P(AI|\text{love})=3/6=0.5$. With Laplace: $P(\text{love}|I)=(4+1)/(10+1000)=5/1010 \approx 0.00495$, $P(AI|\text{love})=(3+1)/(6+1000)=4/1006 \approx 0.00398$. Smoothing prevents zeros for unseen pairs at the cost of slightly lowering observed ones.

PAPER 2 · 50 Marks · Time: 90 min

Q1. (10 marks) Differentiate Markov Chain and HMM. With a 2-state weather example, compute $P(\text{rainy in 2 days} \mid \text{sunny today})$ using a given transition matrix.

Solution: Markov Chain: states observable. HMM: states hidden, only emissions observable. Using $P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$ with state order (Sunny, Rainy): $P^2(\text{Sunny} \rightarrow \text{Rainy}) = 0.7 \cdot 0.3 + 0.3 \cdot 0.6 = 0.21 + 0.18 = 0.39$.

Q2. (10 marks) Explain the three classic problems of an HMM and the algorithms used.

Solution: Evaluation – Forward algorithm computes $P(O \mid \lambda)$. Decoding – Viterbi finds the most likely state sequence. Learning – Baum–Welch (EM) estimates λ from observations.

Q3. (10 marks) Construct truth tables for $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$ and prove the equivalence.

Solution: Both columns yield identical values for all four (P,Q) combinations (T,T)→T, (T,F)→F, (F,T)→T, (F,F)→T. Hence implication and its contrapositive are logically equivalent.

Q4. (10 marks) Translate to predicate logic: (a) Every student passes some exam. (b) No politician is honest. (c) Some teacher loves every student.

Solution: (a) $\forall x (\text{Student}(x) \rightarrow \exists y (\text{Exam}(y) \wedge \text{Passes}(x,y)))$. (b) $\forall x (\text{Politician}(x) \rightarrow \neg \text{Honest}(x))$. (c) $\exists x (\text{Teacher}(x) \wedge \forall y (\text{Student}(y) \rightarrow \text{Loves}(x,y)))$.

Q5. (10 marks) Scenario: A hospital wants an AI system to triage emergency patients. Describe how an expert system + NLP pipeline + HMM could be combined.

Solution: NLP pipeline reads patient complaints (tokenize, NER for symptoms/drugs). Symptoms feed the expert system's knowledge base; forward chaining derives candidate diagnoses with confidence scores. An HMM models disease progression over visits, using observed symptom emissions to predict future severity, helping triage prioritize high-risk patients.

PAPER 3 · 50 Marks · Time: 90 min

Q1. (10 marks) Apply Welsh-Powell on a graph with degrees (A:4, B:3, C:3, D:2, E:2, F:1). Describe each step and final chromatic number.

Solution: Order: A,B,C,D,E,F. Color A=1. Color the largest independent set with color 1 (e.g., A and F if non-adjacent). Color next-largest IS with color 2. Continue. Typical result: 3 colors.

Q2. (10 marks) Prove that with perfect move ordering alpha-beta examines $O(b^{d/2})$ nodes.

Solution: At each level the best child is examined fully (b factor); the rest can be pruned after a single comparison. Total roughly $b^{d/2} \cdot b^{d/2} = b^d$ cut to $b^{d/2}$ effective branching, doubling reachable depth.

Q3. (10 marks) Compare DENDRAL, MYCIN and XCON expert systems.

Solution: DENDRAL inferred molecular structures from mass spectra. MYCIN diagnosed bacterial infections and suggested antibiotics, pioneering certainty factors. XCON configured DEC VAX computer orders, saving DEC ~\$25M/year.

Q4. (10 marks) Build an N-gram model on the toy corpus: 'AI is the future. AI changes the world. Future AI will think.' Compute unigram and bigram probabilities of 'AI'.

Solution: Unigram: $\text{Count}(\text{AI})=3$, total tokens (excluding punctuation) = 11 $\rightarrow P(\text{AI})=3/11 \approx 0.27$. Bigram example $P(\text{AI})=2/3 \approx 0.67$ since 2 of 3 sentences start with AI.

Q5. (10 marks) Using propositional logic, formalize and solve: 'If it rains, the match is cancelled. If the match is cancelled, fans get refunds. It rains. Therefore fans get refunds.'

Solution: Let R=rain, C=cancelled, F=refund. Premises: $R \rightarrow C$, $C \rightarrow F$, R. By Hypothetical Syllogism $R \rightarrow F$; by Modus Ponens with $R \rightarrow F$. Hence fans get refunds. ■

PAPER 4 · 50 Marks · Time: 90 min

Q1. (10 marks) Heuristic: order children for alpha-beta in a chess engine.

Solution: Order: (i) TT-move from transposition table, (ii) captures by MVV-LVA, (iii) killer moves, (iv) history heuristic. Quality of ordering can multiply effective depth.

Q2. (10 marks) Compute trigram $P(\text{world}|\text{l},\text{love})$ given $\text{Count}(\text{l love world})=2$, $\text{Count}(\text{l love})=5$.

Solution: $P(\text{world}|\text{l},\text{love})=2/5=0.4$. With Laplace smoothing and $V=1000$: $(2+1)/(5+1000)=3/1005\approx0.00298$.

Q3. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella: $\delta_2(\text{Rainy})=\max(0.54\cdot0.7, 0.08\cdot0.4)\cdot0.1=0.0378$; $\delta_2(\text{Sunny})=\max(0.54\cdot0.3, 0.08\cdot0.6)\cdot0.8=0.1296$. Best path: Rainy $\rightarrow$ Sunny.

Q4. (10 marks) Scenario: design an AI tutor combining NLP + Expert System + HMM for adaptive testing.

Solution: NLP parses learner answers; expert-system rules diagnose misconceptions; HMM models the learner's evolving knowledge state, choosing the next question to maximize information gain. Together they provide explainable, adaptive tutoring.

Q5. (10 marks) Solve graph coloring on a 6-vertex cycle C_6 .

Solution: C_6 is bipartite — vertices alternate. $\chi(C_6)=2$ using Red/Blue alternating around the cycle.

PAPER 5 · 50 Marks · Time: 90 min

Q1. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q2. (10 marks) List 6 NLP preprocessing steps and give examples for each.

Solution: 1) Lowercasing 'The'→'the'. 2) Tokenization. 3) Stop-word removal. 4) Stemming/Lemmatization. 5) POS tagging. 6) NER. Each progressively reduces noise and enriches semantic information.

Q3. (10 marks) Find steady-state distribution for $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$.

Solution: Solve $\pi P = \pi$ with $\pi_1 + \pi_2 = 1$: $\pi_1 = 0.2\pi_1 + 0 \dots$ actually $\pi_1 \cdot 0.6 + \pi_2 \cdot 0.2 = \pi_1 \Rightarrow 0.4\pi_1 = 0.2\pi_2 \Rightarrow \pi_2 = 2\pi_1$. With $\pi_1 + \pi_2 = 1$: $\pi_1 = 1/3$, $\pi_2 = 2/3$.

Q4. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q5. (10 marks) Solve graph coloring on a 6-vertex cycle C_6 .

Solution: C_6 is bipartite — vertices alternate. $\chi(C_6) = 2$ using Red/Blue alternating around the cycle.

PAPER 6 · 50 Marks · Time: 90 min

Q1. (10 marks) Scenario: design an AI tutor combining NLP + Expert System + HMM for adaptive testing.

Solution: NLP parses learner answers; expert-system rules diagnose misconceptions; HMM models the learner's evolving knowledge state, choosing the next question to maximize information gain. Together they provide explainable, adaptive tutoring.

Q2. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella:
 $\delta_2(\text{Rainy})=\max(0.54 \cdot 0.7, 0.08 \cdot 0.4) \cdot 0.1=0.0378$; $\delta_2(\text{Sunny})=\max(0.54 \cdot 0.3, 0.08 \cdot 0.6) \cdot 0.8=0.1296$. Best path: Rainy $\rightarrow$ Sunny.

Q3. (10 marks) Solve graph coloring on a 6-vertex cycle C_6 .

Solution: C_6 is bipartite — vertices alternate. $\chi(C_6)=2$ using Red/Blue alternating around the cycle.

Q4. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim \frac{1}{3}$ of nodes are visited vs full minimax.

Q5. (10 marks) Predicate-logic formalization of: 'Some students attend every lecture.'

Solution: $\exists x (\text{Student}(x) \wedge \forall y (\text{Lecture}(y) \rightarrow \text{Attends}(x,y)))$.

PAPER 7 · 50 Marks · Time: 90 min

Q1. (10 marks) Build a small bigram model on 'data is the new oil; data drives AI'.

Solution: Bigrams: (data,is)=1, (is,the)=1, (the,new)=1, (new,oil)=1, (data,drives)=1, (drives,AI)=1.
 $P(\text{drives}|\text{data})=1/2$, $P(\text{is}|\text{data})=1/2$.

Q2. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q3. (10 marks) List 6 NLP preprocessing steps and give examples for each.

Solution: 1) Lowercasing 'The'→'the'. 2) Tokenization. 3) Stop-word removal. 4) Stemming/Lemmatization. 5) POS tagging. 6) NER. Each progressively reduces noise and enriches semantic information.

Q4. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically ~ $\frac{1}{3}$ of nodes are visited vs full minimax.

Q5. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$.

PAPER 8 · 50 Marks · Time: 90 min

Q1. (10 marks) Build a small bigram model on 'data is the new oil; data drives AI'.

Solution: Bigrams: (data,is)=1, (is,the)=1, (the,new)=1, (new,oil)=1, (data,drives)=1, (drives,AI)=1.
 $P(\text{drives}|\text{data})=1/2$, $P(\text{is}|\text{data})=1/2$.

Q2. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in [0,9].

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$.
Typically ~1/3 of nodes are visited vs full minimax.

Q3. (10 marks) Scenario: design an AI tutor combining NLP + Expert System + HMM for adaptive testing.

Solution: NLP parses learner answers; expert-system rules diagnose misconceptions; HMM models the learner's evolving knowledge state, choosing the next question to maximize information gain. Together they provide explainable, adaptive tutoring.

Q4. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q5. (10 marks) Predicate-logic formalization of: 'Some students attend every lecture.'

Solution: $\exists x (\text{Student}(x) \wedge \forall y (\text{Lecture}(y) \rightarrow \text{Attends}(x,y)))$.

PAPER 9 · 50 Marks · Time: 90 min

Q1. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q2. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim 1/3$ of nodes are visited vs full minimax.

Q3. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q4. (10 marks) Explain Resolution in propositional logic with an example.

Solution: From clauses $(P \vee Q)$ and $(\neg P \vee R)$ infer $(Q \vee R)$. Repeated resolution on CNF eventually produces empty clause when statement is contradictory — proof by refutation.

Q5. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella:
 $\delta_2(\text{Rainy}) = \max(0.54 \cdot 0.7, 0.08 \cdot 0.4) \cdot 0.1 = 0.0378$; $\delta_2(\text{Sunny}) = \max(0.54 \cdot 0.3, 0.08 \cdot 0.6) \cdot 0.8 = 0.1296$. Best path: Rainy $\rightarrow$ Sunny.

PAPER 10 · 50 Marks · Time: 90 min

Q1. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in [0,9].

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim \frac{1}{3}$ of nodes are visited vs full minimax.

Q2. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q3. (10 marks) Solve graph coloring on a 6-vertex cycle C_6 .

Solution: C_6 is bipartite — vertices alternate. $\chi(C_6)=2$ using Red/Blue alternating around the cycle.

Q4. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$.

Q5. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

PAPER 11 · 50 Marks · Time: 90 min

Q1. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q2. (10 marks) Build a small bigram model on 'data is the new oil; data drives AI'.

Solution: Bigrams: (data,is)=1, (is,the)=1, (the,new)=1, (new,oil)=1, (data,drives)=1, (drives,AI)=1. $P(\text{drives}|\text{data})=1/2$, $P(\text{is}|\text{data})=1/2$.

Q3. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q4. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$.

Q5. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in [0,9].

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically ~1/3 of nodes are visited vs full minimax.

PAPER 12 · 50 Marks · Time: 90 min

Q1. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q2. (10 marks) Find steady-state distribution for $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$.

Solution: Solve $\pi P = \pi$ with $\pi_1 + \pi_2 = 1$: $\pi_1 = 0.2\pi_1 + 0.2\pi_2$... actually $\pi_1 \cdot 0.6 + \pi_2 \cdot 0.2 = \pi_1 \Rightarrow 0.4\pi_1 = 0.2\pi_2 \Rightarrow \pi_2 = 2\pi_1$. With $\pi_1 + \pi_2 = 1$: $\pi_1 = 1/3$, $\pi_2 = 2/3$.

Q3. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim \frac{1}{3}$ of nodes are visited vs full minimax.

Q4. (10 marks) Scenario: design an AI tutor combining NLP + Expert System + HMM for adaptive testing.

Solution: NLP parses learner answers; expert-system rules diagnose misconceptions; HMM models the learner's evolving knowledge state, choosing the next question to maximize information gain. Together they provide explainable, adaptive tutoring.

Q5. (10 marks) Build a small bigram model on 'data is the new oil; data drives AI'.

Solution: Bigrams: (data,is)=1, (is,the)=1, (the,new)=1, (new,oil)=1, (data,drives)=1, (drives,AI)=1. $P(\text{drives}|\text{data})=1/2$, $P(\text{is}|\text{data})=1/2$.

PAPER 13 · 50 Marks · Time: 90 min

Q1. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella:
 $\delta_2(\text{Rainy}) = \max(0.54 \cdot 0.7, 0.08 \cdot 0.4) \cdot 0.1 = 0.0378$; $\delta_2(\text{Sunny}) = \max(0.54 \cdot 0.3, 0.08 \cdot 0.6) \cdot 0.8 = 0.1296$. Best path: Rainy $\rightarrow$ Sunny.

Q2. (10 marks) Graph coloring on Petersen graph — find χ .

Solution: Petersen graph: $\chi=3$. Bipartite test fails (odd cycles present) so $\chi \geq 3$; a valid 3-coloring exists.

Q3. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim \frac{1}{3}$ of nodes are visited vs full minimax.

Q4. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$.

Q5. (10 marks) Predicate-logic formalization of: 'Some students attend every lecture.'

Solution: $\exists x (\text{Student}(x) \wedge \forall y (\text{Lecture}(y) \rightarrow \text{Attends}(x,y)))$.

PAPER 14 · 50 Marks · Time: 90 min

Q1. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{combine} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q2. (10 marks) Apply Alpha-Beta on a tree of depth 4 branching factor 3 with leaves randomly in $[0,9]$.

Solution: Use best-first ordering: explore leftmost MAX child fully, prune sibling MIN children when $\alpha \geq \beta$. Typically $\sim \frac{1}{3}$ of nodes are visited vs full minimax.

Q3. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q4. (10 marks) List 6 NLP preprocessing steps and give examples for each.

Solution: 1) Lowercasing 'The' $\rightarrow$ 'the'. 2) Tokenization. 3) Stop-word removal. 4) Stemming/Lemmatization. 5) POS tagging. 6) NER. Each progressively reduces noise and enriches semantic information.

Q5. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

PAPER 15 · 50 Marks · Time: 90 min

Q1. (10 marks) Build a small bigram model on 'data is the new oil; data drives AI'.

Solution: Bigrams: (data,is)=1, (is,the)=1, (the,new)=1, (new,oil)=1, (data,drives)=1, (drives,AI)=1.
 $P(\text{drives}|\text{data})=1/2$, $P(\text{is}|\text{data})=1/2$.

Q2. (10 marks) Heuristic: order children for alpha-beta in a chess engine.

Solution: Order: (i) TT-move from transposition table, (ii) captures by MVV-LVA, (iii) killer moves, (iv) history heuristic. Quality of ordering can multiply effective depth.

Q3. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q4. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q5. (10 marks) Explain Resolution in propositional logic with an example.

Solution: From clauses $(P \vee Q)$ and $(\neg P \vee R)$ infer $(Q \vee R)$. Repeated resolution on CNF eventually produces empty clause when statement is contradictory — proof by refutation.

PAPER 16 · 50 Marks · Time: 90 min

Q1. (10 marks) Graph coloring on Petersen graph — find χ .

Solution: Petersen graph: $\chi=3$. Bipartite test fails (odd cycles present) so $\chi \geq 3$; a valid 3-coloring exists.

Q2. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q3. (10 marks) List 6 NLP preprocessing steps and give examples for each.

Solution: 1) Lowercasing 'The'→'the'. 2) Tokenization. 3) Stop-word removal. 4) Stemming/Lemmatization. 5) POS tagging. 6) NER. Each progressively reduces noise and enriches semantic information.

Q4. (10 marks) Difference between syntactic and semantic analysis in NLP.

Solution: Syntactic analysis (parsing) extracts grammatical structure — POS, parse trees. Semantic analysis assigns meaning — word-sense disambiguation, semantic-role labelling, knowledge-graph linking.

Q5. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

PAPER 17 · 50 Marks · Time: 90 min

Q1. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q2. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_{\text{combine}} = CF_1 + CF_2(1 - CF_1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q3. (10 marks) Scenario: Auto-translate 'I love mangoes' to French using statistical MT in 3 stages.

Solution: 1) Tokenize. 2) Align to French training pairs (phrase table). 3) Decode with language model: 'J'aime les mangues'. Modern systems use Transformers (seq2seq with attention).

Q4. (10 marks) Compute trigram $P(\text{world}|\text{I}, \text{love})$ given $\text{Count}(\text{I love world})=2$, $\text{Count}(\text{I love})=5$.

Solution: $P(\text{world}|\text{I}, \text{love}) = 2/5 = 0.4$. With Laplace smoothing and $V=1000$: $(2+1)/(5+1000) = 3/1005 \approx 0.00298$.

Q5. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x))$.

PAPER 18 · 50 Marks · Time: 90 min

Q1. (10 marks) Predicate-logic formalization of: 'Some students attend every lecture.'

Solution: $\exists x (\text{Student}(x) \wedge \forall y (\text{Lecture}(y) \rightarrow \text{Attends}(x,y)))$.

Q2. (10 marks) Heuristic: order children for alpha-beta in a chess engine.

Solution: Order: (i) TT-move from transposition table, (ii) captures by MVV-LVA, (iii) killer moves, (iv) history heuristic. Quality of ordering can multiply effective depth.

Q3. (10 marks) Graph coloring on Petersen graph — find χ .

Solution: Petersen graph: $\chi=3$. Bipartite test fails (odd cycles present) so $\chi \geq 3$; a valid 3-coloring exists.

Q4. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q5. (10 marks) Compute trigram $P(\text{world}|\text{l},\text{love})$ given $\text{Count}(\text{l love world})=2$, $\text{Count}(\text{l love})=5$.

Solution: $P(\text{world}|\text{l},\text{love})=2/5=0.4$. With Laplace smoothing and $V=1000$:
 $(2+1)/(5+1000)=3/1005 \approx 0.00298$.

PAPER 19 · 50 Marks · Time: 90 min

Q1. (10 marks) Describe the role of certainty factors in MYCIN.

Solution: Certainty factors (CF in $[-1,1]$) quantify confidence in rules. Combined via: $CF_combine = CF1 + CF2(1 - CF1)$ for same-sign positives. Allows uncertain reasoning without full Bayesian machinery.

Q2. (10 marks) List 6 NLP preprocessing steps and give examples for each.

Solution: 1) Lowercasing 'The'→'the'. 2) Tokenization. 3) Stop-word removal. 4) Stemming/Lemmatization. 5) POS tagging. 6) NER. Each progressively reduces noise and enriches semantic information.

Q3. (10 marks) Translate: 'Every prime number greater than 2 is odd.'

Solution: $\forall x ((\text{Prime}(x) \wedge x > 2) \rightarrow \text{Odd}(x)).$

Q4. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella: $\delta_2(\text{Rainy}) = \max(0.54 \cdot 0.7, 0.08 \cdot 0.4) \cdot 0.1 = 0.0378$; $\delta_2(\text{Sunny}) = \max(0.54 \cdot 0.3, 0.08 \cdot 0.6) \cdot 0.8 = 0.1296$. Best path: Rainy → Sunny.

Q5. (10 marks) Scenario: design an AI tutor combining NLP + Expert System + HMM for adaptive testing.

Solution: NLP parses learner answers; expert-system rules diagnose misconceptions; HMM models the learner's evolving knowledge state, choosing the next question to maximize information gain. Together they provide explainable, adaptive tutoring.

PAPER 20 · 50 Marks · Time: 90 min

Q1. (10 marks) Heuristic: order children for alpha-beta in a chess engine.

Solution: Order: (i) TT-move from transposition table, (ii) captures by MVV-LVA, (iii) killer moves, (iv) history heuristic. Quality of ordering can multiply effective depth.

Q2. (10 marks) Compute trigram $P(\text{world}|\text{l},\text{love})$ given $\text{Count}(\text{l love world})=2$, $\text{Count}(\text{l love})=5$.

Solution: $P(\text{world}|\text{l},\text{love})=2/5=0.4$. With Laplace smoothing and $V=1000$: $(2+1)/(5+1000)=3/1005\approx 0.00298$.

Q3. (10 marks) Explain Resolution in propositional logic with an example.

Solution: From clauses $(P\vee Q)$ and $(\neg P\vee R)$ infer $(Q\vee R)$. Repeated resolution on CNF eventually produces empty clause when statement is contradictory — proof by refutation.

Q4. (10 marks) Truth-table-prove De Morgan's Law: $\neg(P\wedge Q) \equiv \neg P\vee\neg Q$.

Solution: Construct 4-row table; both expressions evaluate identically: F,T,T,T. Hence equivalent.

Q5. (10 marks) Run Viterbi for 2 days with given HMM (use values from Chapter 7).

Solution: Day 1 best=Rainy ($\alpha=0.54$). Day 2 observation = NoUmbrella: $\delta_2(\text{Rainy})=\max(0.54\cdot 0.7, 0.08\cdot 0.4)\cdot 0.1=0.0378$; $\delta_2(\text{Sunny})=\max(0.54\cdot 0.3, 0.08\cdot 0.6)\cdot 0.8=0.1296$. Best path: Rainy $\rightarrow$ Sunny.

EXAM DAY SURVIVAL GUIDE

Time Management (90-min, 50-mark paper)

Time	Activity
0–5 min	Read all questions, mark easy ones with ★
5–70 min	Solve easy → medium → hard in that order
70–85 min	Attempt the hardest question
85–90 min	Review diagrams, units, numbering

Memory Tricks

- **'Bill Forwards Goals Backwards'** — Forward chaining is data-driven, Backward chaining is goal-driven.
- **'AB → MM'** — Alpha-Beta optimizes Minimax.
- **'HMM = Hidden Music Maker'** — you hear the music (observations) but not the musician (state).
- **' $\forall$ looks like A (All), $\exists$ looks like E (Exists)'**.
- **' χ counts colors, like χ mas trees'** — chromatic number stores in your head.

Top 10 Common Mistakes

- Confusing Markov Chain with HMM.
- Forgetting to check $\alpha \geq \beta$ BEFORE recursion in alpha-beta.
- Mixing stemming with lemmatization.
- Forgetting Laplace smoothing in N-gram numerical questions.
- Drawing expert-system architecture with missing 'Explanation Facility' box.
- Reversing the direction of implication arrow.
- Using $\forall$ where $\exists$ is needed (and vice versa).
- Forgetting that rows of a Markov matrix sum to 1.
- Coloring an adjacent pair with the same color in CSP problems.
- Writing prose answers for diagram questions — always include the figure.

Final Pep Talk

You now hold a complete AI academy in one book. Every concept has a story, a definition, a diagram and a solved problem. Practice 5 papers under timed conditions, revisit your weak chapters, and walk into the exam with the confidence of someone who has been taught by a tutor, a textbook and a visual tutor — all combined.

Good luck, future AI engineer. The world needs you. ■